

Sushant S. Mahajan

Solar Astrophysicist • Data Scientist • AI for Science
solarmagnetism.org ORCID: 0000-0003-1753-8002

SUMMARY

Research scientist and computational physicist building data-driven systems for solar physics, scientific inference, time-series analysis, scientific AI systems, and space-weather applications.

EXPERIENCE

Research Scientist, W. W. Hansen Experimental Physics Laboratory, Stanford University 2024–present

- Responsible for developing an agentic AI-powered JSOC Data Commons server for public query and visualization of SDO data.
- Lead research on solar dynamics and large-scale flows from HMI/SDO observations.
- Responsible for producing new electric and velocity field-based metadata parameters from SDO/HMI data.
- Collaborated across research, software, and data management teams on mission-facing analysis.

Postdoctoral Fellow, W. W. Hansen Experimental Physics Laboratory, Stanford University 2022–2024

- Published first-author work on differential rotation, torsional oscillation, and meridional flow.
- Helped deliver the Stanford Solar Observatories Group data-sharing effort recognized in 2026.
- Contributed to forecasting-oriented work on rare events and solar flare data.

Solar Physics Postdoctoral Fellow, Institute for Astronomy, University of Hawai'i 2019–2022

- Built analysis pipelines for active-region inflows, solar-cycle trends, and helioseismic flow inference.
- Contributed to forecasting-oriented work on rare events and solar flare data.
- Developed workflows for large-scale solar magnetogram analysis and space-weather-relevant diagnostics.

SELECTED IMPACT

- NASA Silver Group Achievement Award for JSOC flood recovery efforts 2025
- Stanford University Libraries Data Sharing Prize, Stanford Solar Observatories Group 2026
- Editor's Choice, *Solar Physics*, for *The Sun's Large-Scale Flows I* 2024
- Prepared and published the SWAN-SF benchmark dataset used in solar-flare AI work 2020
- Presentation awards: Best Young Presenter Award (IAU Symposium 340), Best Young Scientist Poster Award (IAU Symposium 335), Honorary Mention (SPD meeting) 2018–2016

SELECTED TALKS & MEDIA

“Unveiling the Sun: Exploring the Wonders of Solar Astrophysics,” KIPAC public lecture, Stanford University 2023

“The Influence of Active Regions on Plasma Flows Inside the Sun,” UCAR/HAO colloquium 2023

“Unveiling the Solar Poles: Addressing Projection Artifacts and Revealing Circumpolar Features in HMI Magnetograms,” AAS/SPD 246, Anchorage, AK 2025

KRON4 interview on the Northern Lights 2024

NBC Bay Area appearance on the Northern Lights 2024

ABC7 Bay Area appearance on the Northern Lights [2024](#)

Quoted in *San Francisco Chronicle* on California aurora visibility [2024](#)

Quoted in *San Francisco Chronicle* on solar maximum and renewed aurora prospects [2024](#)

Stanford Report, “Four Questions with a Scientist” on the 2024 total solar eclipse [2024](#)

SELECTED PUBLICATIONS

Ding, B., Zhao, J., Chen, R., Waidele, M., **Mahajan, S. S.**, and Vesa, O.
Characterizing the Observational Properties of the Sun’s High-latitude $m = 1$ Inertial Mode. *Astrophysical Journal* [2025](#)

Mahajan, S. S., Upton, L. A., Antia, H. M., Basu, S., DeRosa, M. L., Hess Webber, S. A., et al.
The Sun’s Large-Scale Flows I: Measurements of Differential Rotation & Torsional Oscillation. *Solar Physics* 299, 38 [2024](#)

Mahajan, S. S., Sun, X., and Zhao, J.
Removal of Active Region Inflows Reveals a Weak Solar Cycle Scale Trend in the Near-Surface Meridional Flow. *Astrophysical Journal* 950, 63 [2023](#)

Dash, S., DeRosa, M. L., Dikpati, M., Sun, X., **Mahajan, S. S.**, Liu, Y., and Hoeksema, J. T.
Ensemble Kalman Filter Data Assimilation into the Surface Flux Transport Model to Infer Surface Flows. *Astrophysical Journal* [2024](#)

For full and current publication list see [Google Scholar](#).

SKILLS

Python, MATLAB, Fortran, bash, OpenMP, MPI, C++, CUDA, scientific computing, time-series analysis, machine learning workflows, data analysis, visualization, and technical writing.

EDUCATION

Ph.D. in Astronomy, Georgia State University [2019](#)
M.S. in Physics, Georgia State University [2017](#)
M.Tech. in Engineering Physics, IIT (BHU) [2014](#)